

Store Sales Forecasting and demand prediction

Project planning phase

1. Project Proposal

1.1 Project Overview

This project addresses the Store Sales for Corporación Favorita, a large Ecuadorian-based grocery retailer. Accurate demand prediction is critical for inventory optimization, supply chain management, and revenue planning in the highly competitive retail sector.

The dataset have sales records from January 2013 to August 2017, covering 54 retail stores, 33 product families, and approximately 3 million training observations. Additional contextual data includes oil prices, national holidays and events, store metadata, and promotional indicators.

1.2 Problem Statement

Retail chains face persistent challenges with stock outs, overstock, and promotional planning due to the complexity of demand patterns across product categories, store locations, and time. The goal is to build a machine learning-based time series forecasting pipeline that predicts daily unit sales for each store-product family combination over a future horizon of 15 days.

1.3 Project Objectives

- Develop a robust time series forecasting model.
- Perform comprehensive Exploratory Data Analysis (EDA) across stores, product families, time, and external factors.
- Engineer high-quality features from temporal patterns, lag variables, oil prices, promotions, and holiday events.

- Evaluate and compare multiple modeling approaches: baseline statistical models, gradient boosting (LightGBM/XGBoost), and deep learning (LSTM/Transformer).
- Deliver reproducible, well-documented code, notebooks, and formal reports aligned with academic milestones.

1.4 Project Scope

In Scope:

- Data ingestion and preprocessing of all seven provided files (train.csv, test.csv, stores.csv, oil.csv, holidays_events.csv, transactions.csv, sample_submission.csv).
- Exploratory Data Analysis with visualizations across all data dimensions.
- Feature engineering: temporal, lag, rolling statistics, promotions, holiday encoding, oil price impact.
- Model training, validation, and hyper parameter tuning.
- Final prediction generation.
- Project documentation: proposal, plan, risk register, KPI tracking, and milestone reports.

Out of Scope:

- Real-time or production deployment of the forecasting system.
- External data sources beyond those provide.
- Mobile or web application development.

1.5 Dataset Summary

File	Rows	Key Columns	Purpose
train.csv	~3,000,888	date, store_nbr,	Model training

		family, sales, onpromotion	
test.csv	28,512	date, store_nbr, family, onpromotion	Prediction target
stores.csv	54	store_nbr, city, state, type, cluster	Store metadata
oil.csv	1,218	date, dcoilwtico	External factor
holidays_events.csv	350	date, type, locale, locale_name, transferred	Event encoding
transactions.csv	83,488	date, store_nbr, transactions	Store traffic

2. Project Plan

2.1 Gantt Chart — 8-Week Timeline

The following Gantt chart outlines the planned timeline for all project phases across an 8-week period.

Phase / Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Data Collection & Setup								
EDA & Visualization								
Feature Engineering								
Baseline Modeling								
Advanced Modeling								
Hyperparameter Tuning								
Final Submission & Report								

2.2 Milestones & Deliverables

M#	Milestone	Deliverables	Target Week	Status
M1	Project Initiation	Proposal, project plan, risk register, KPIs	Week 1	Complete
M2	Data Understanding	EDA notebook, visualizations, insights report	Week 3	Complete
M3	Feature Engineering	Feature set, engineered dataset, feature importance analysis	Week 4	In progress
M4	Baseline Models	Baseline RMSLE, model comparison table	Week 5	Pending
M5	Advanced Modeling	LightGBM/XGBoost tuned models, RMSLE < 0.45	Week 6	Pending
M6	Final Submission	submission CSV, final report, code repository	Week 8	Pending

2.3 Resource Allocation

Resource	Type	Usage / Tooling	Allocation
Project Team	Human	Python, Jupyter,	Full-time 8 weeks

		GitHub collaboration	
Compute Environment	Hardware/Cloud	Google Colab	As needed
Libraries & Frameworks	Software	pandas, scikit-learn, LightGBM, XGBoost, TensorFlow, matplotlib	Throughout
Version Control	Software	GitHub repository with branching strategy	Throughout
Documentation Tools	Software	Microsoft Word, Jupyter nbconvert, LaTeX	Per milestone

3. Task Assignment & Roles

The project team is structured around four core roles, each with defined responsibilities across the project lifecycle. Responsibilities may be shared during peak workload phases to ensure timely delivery.

Role	Name / Owner	Responsibilities	Primary Phases
Project Manager	Team Lead	Overall coordination, timeline tracking, stakeholder communication, milestone reporting, risk monitoring	All phases
Data Engineer	Data Specialist	Data ingestion, cleaning, merging datasets, handling missing values, train/test splitting, pipeline automation	Weeks 1–3
EDA & Visualization Analyst	Analyst	Statistical analysis, distribution plots, correlation heatmaps, time series decomposition,	Weeks 2–3

		holiday/event impact analysis	
ML / Feature Engineer	ML Engineer	Lag features, rolling windows, cyclical encoding, oil price integration, promotion flags, model training & evaluation	Weeks 3–7
Report & Documentation Lead	Documentation Specialist	Milestone Word reports, notebook formatting, submission documentation, code commenting, README maintenance	Weeks 1, 3, 5, 8

3.1 Detailed Task Assignments by Phase

Phase	Tasks	Assigned Role(s)
Data Preparation	Load all 6 CSVs, merge on store_nbr and date, handle ~1,200 missing oil values via forward-fill	Data Engineer
EDA	Sales trends by store/family, seasonal decomposition, promotion impact, holiday effects, oil	EDA Analyst + ML Engineer

	correlation analysis	
Feature Engineering	Lag-1/7/14/28 sales, 7/14-day rolling mean & std, day-of-week, month, payday flags, earthquake dummy (2016-04), store cluster encoding	ML / Feature Engineer
Baseline Modeling	Moving average, seasonal naïve, linear regression baselines; compute RMSLE for comparison	ML Engineer
Advanced Modeling	LightGBM with time-series cross-validation, XGBoost, optional LSTM; hyperparameter tuning via Optuna/GridSearch	ML / Feature Engineer
Reporting	Compile milestone documents, embed visualizations, executive summary, KPI dashboard update	Report Lead + Project Manager

4. Risk Assessment & Mitigation Plan

The following risk register identifies potential threats to the project's success, rated by impact (effect on objectives if realized) and probability (likelihood of occurrence). Mitigation strategies are pre planned actions to reduce or eliminate each risk.

ID	Risk Name	Description	Impact	Prob.	Mitigation Strategy
R01	Data Quality Issues	Missing oil prices (~43 missing dates), zero-sales days for new stores, outliers during 2016 earthquake	High	Medium	Interpolate oil prices; apply zero-sales masking for store opening periods; cap outlier sales at 99.9th percentile
R02	Overfitting to Training Data	Model learns training-period patterns that don't generalize to the 15-day	High	Medium	Time-series cross-validation with 5-fold walk-forward

		forecast horizon			splits; monitor train vs. validation RMSLE gap
R0 3	Feature Leakage	Inadvertently using future data in lag/rolling features during training	High	Low	Strict date-based splitting; shift all lag features correctly; unit tests on feature creation pipeline
R0 4	Computational Resource Limits	Training LightGBM on ~3M rows may be slow on local hardware; GPU notebooks have runtime limits	Medium	Medium	cache intermediate feature datasets as parquet
R0 5	Holiday & Event Complexity	Ecuador has national, regional, and local	Medium	Medium	Map holidays by store city;

		holidays; misclassification leads to systematic forecast errors			include transferred holiday flags; create 'holiday window' ± 2 day features
R06	Team Coordination Delays	Merge conflicts in Jupyter notebooks; asynchronous work causing integration issues	Medium	Low	Separate notebooks per task; use nbstripout for cleaner git diffs; weekly sync meetings
R07	Model Underperformance	Final RMSLE exceeds target threshold (> 0.50) due to inherent data noise or insufficient features	Medium	Medium	Ensemble stacking of LightGBM + XGBoost; add store-level scaling; analyze residual

					patterns for remaining signal
R08	Scope Creep	Additional experiments (deep learning, external datasets) consuming time beyond planned schedule	Low	Medium	Freeze core scope at Week 4; additional experiments only if core objectives are met ahead of schedule

5. Key Performance Indicators (KPIs)

ID	KPI Name	Description	Target	Frequency	Corrective Action
KPI-01	Model Accuracy (RMSLE)	Quantifies prediction accuracy evaluation metric (Root Mean Squared Logarithmic Error)	RMSLE $\leq$ 0.45 on public leaderboard	Computed after each model training run	RMSLE > 0.60 after baseline = re-engineer features
KPI-02	Feature Engineering Coverage	% of available external signals incorporated as model features	$\geq$ 85% of identified signals used	Reviewed at Milestone 3	Add missing features before modeling phase begins
KPI-03	Data Completeness Rate	% of training rows with no missing values after preprocessing	$\geq$ 99.5% complete rows	After data preparation phase	Additional imputation strategies applied
KPI-04	Milestone Delivery Rate	% of project milestones delivered on or before their	100% on-time delivery	End of each milestone week	Buffer week and scope reduction if behind schedule

		target date			
KPI-05	Validation vs. Training RMSLE Gap	Difference between training and validation RMSLE — measures overfitting	Gap < 0.05	After each model training	Apply stronger regularization, reduce feature collinearity
KPI-06	Code Coverage & Reproducibility	% of critical pipeline steps documented and reproducible via a single script/notebook run	100% reproducible pipeline	Before final submission	Add missing documentation; test fresh environment run
KPI-07	EDA Insight Quality	Number of actionable insights derived from EDA that informed feature engineering decisions	≥ 10 documented insights	After Milestone 2 EDA report	Revisit EDA; analyze additional data slices (by city, cluster)
